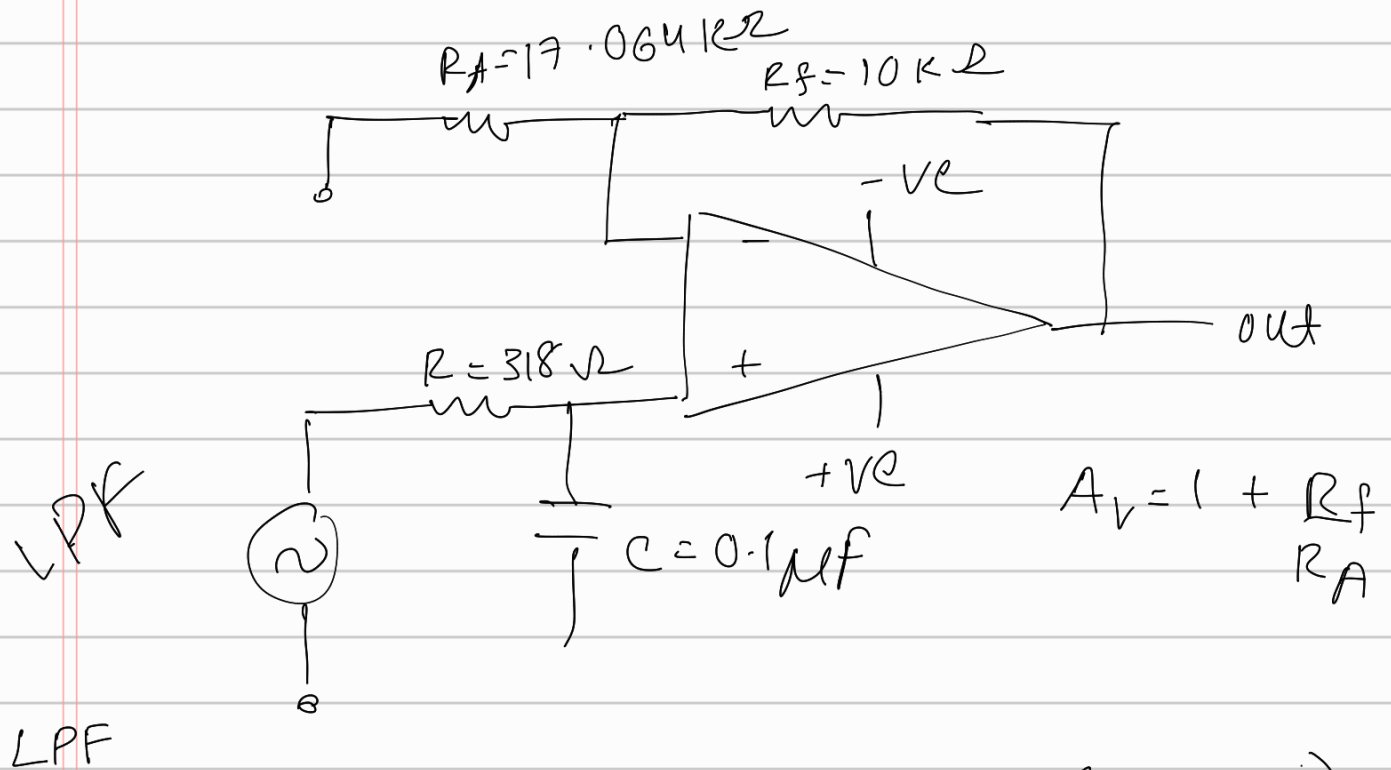


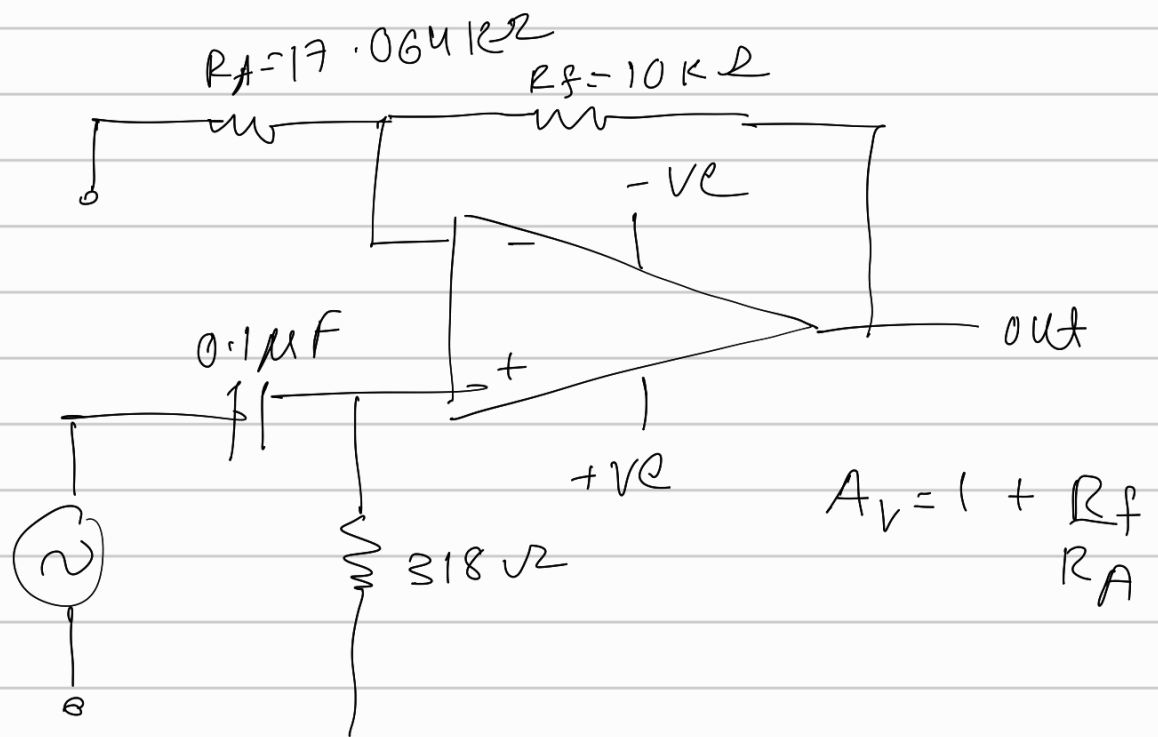
Objective: Design 1st order active butter worth filters: LPF & HPF



(db = 20 \log A_v)

I / P (5V _{pp})	V _o	A _v	db
100 Hz	8.6	1.72	4.71
200 Hz	8.6	1.72	4.71
300 Hz	8.6	1.72	4.71
400 Hz	8.6	1.72	4.71
500 Hz	8.6	1.72	4.71
1k	8.4	1.68	4.50
2k	8.2	1.64	4.29
3k	7.8	1.56	3.86
4k	7.4	1.48	3.40
5k	7.0	1.4	2.92
6k	6.6	1.32	2.41
7k	5.8	1.16	1.28
8k	5.4	1.08	0.668
9k	5.2	1.04	0.34
10k	4.4	0.88	-1.1

HPF



HPF

($\text{dB} = 20 \log A_v$)

I/P (Sv_{pp})	V_o (V)	A_v	dB
100 Hz	0.16	0.032	-29.89
200 Hz	0.32	0.064	-23.87
300 Hz	0.4	0.08	-21.93
400 Hz	0.48	0.096	-20.35
500 Hz	0.56	0.112	-19.01
1k	1.04	0.208	-13.63
2k	2	0.4	-7.95
4k	3.52	0.704	-3.04
6k	4.62	0.924	-0.68
8k	5.44	1.08	-0.66
10k	6	1.2	1.56
15k	6.01	1.21	1.65
20k	6.02	1.19	1.51
30k	6.07	1.22	1.727